

Q. Count the total number of divisors of a number.

$$\hookrightarrow n = 24, 36$$

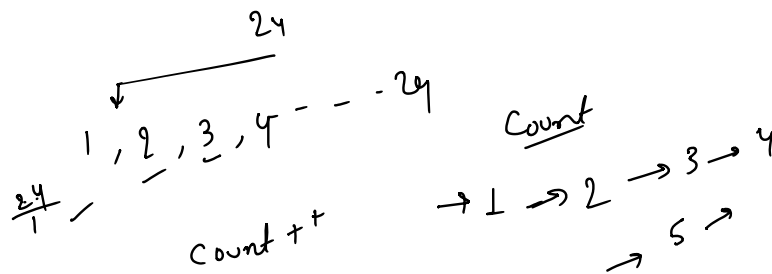
int count = 0;

for (int i = 1; i <= n; i++) {

if (n % i == 0) count++;

}

b



$\sqrt{n}$

$$\hookrightarrow n = 36.$$

{1, 2, 3, 4, ..., 36}

$$\frac{36}{1} \rightarrow (1, 36) \rightarrow 2$$

$$(2, 18) \rightarrow 2$$

$$\frac{36}{2} \rightarrow (3, 12) \rightarrow 2$$

$$\frac{36}{3} \rightarrow (4, 9) \rightarrow 2$$

$$(6, 6) \rightarrow 1$$

$$(9, 4) \rightarrow x$$

$$\sqrt{36} = 6$$

$$\sqrt{36} = 6$$

```

for (int i = 1; i <= sqrt(n); i++) {
    if (n % i == 0) {
        if (n / i == i) count++;
        else count += 2;
    }
}

```

Q. write the code to find HCF/GCD of two Numbers.

Sol:- HCF (9, 36)

9 → 1, 3, (9)

36 → 1, 2, 3, 4, 6, (9), 12, 18, 36

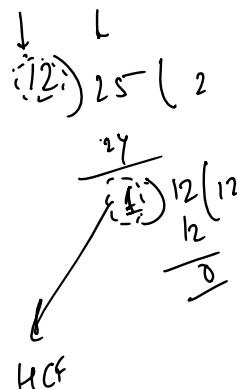
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int hcf = 1;
for (int i = 1; i <= min(a, b); i++) {
    if (a % i == 0 && b % i == 0) {
        hcf = i;
    }
}

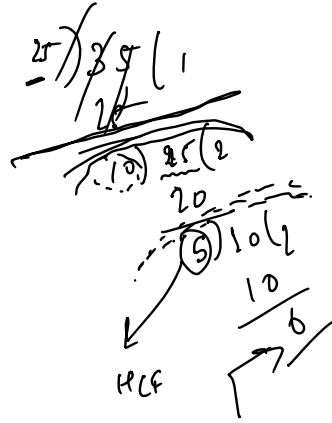
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1 → 10000

HCF (12, 25)



HCF(25, 35)



→ HCF(25, 35)
→ HCF(10, 25)
→ HCF(5, 10)

HCF(25, 35)
= HCF(5, 10)

int a = 25, b = 35;

while (a > 0 && b > 0) {

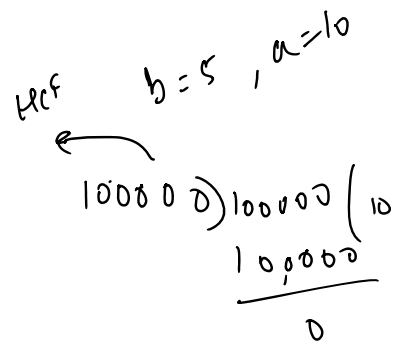
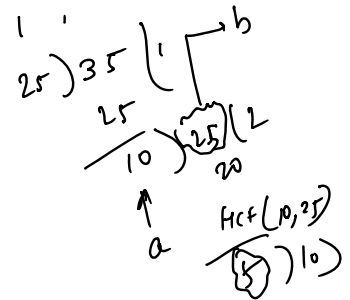
if (a > b) a = a % b;

else b = b % a;

}

if (a == b) sout(b)

else sout(a);



a = 10 0 0 0 0 0

b = 10 0 0 0 0

Q. Find the LCM of two numbers.